

Improving Information Retrieval: A Hybrid of BM25 and Latent Semantic Analysis

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Abstract. We improve the baseline TF-IDF Vector Space Model for the Cranfield collection using three methods: Latent Semantic Analysis (LSA), Okapi BM25, and a hybrid of both. All systems use the same preprocessing and evaluation pipeline for fair comparison. We tune the model parameters using parameter sweeps.

Experiments on 225 Cranfield queries show that the hybrid model performs best and significantly improves over the baseline on most evaluation metrics. LSA and BM25 each solve different weaknesses of the baseline, such as vocabulary mismatch, long-document bias, and word ambiguity, which explains why combining them gives better results.

Keywords: Information Retrieval · Vector Space Model · Latent Semantic Analysis · BM25 · Hybrid Retrieval · Cranfield Collection · Hypothesis Testing

1 Introduction

The goal of this project is to rank documents based on how well they answer a query. We use the classical TF-IDF Vector Space Model, which is simple, fast, and widely used as a baseline for information retrieval systems.

Using the Cranfield collection of 1400 documents and 225 queries, we build and evaluate a baseline retrieval system. The system achieves a Mean Average Precision (MAP) of 0.3122 at $k = 10$. While it works well for simple queries, it struggles when different words express the same meaning or when unrelated documents share common terms.

1.1 Improving the Information Retrieval System

This report improves the TF-IDF baseline using two classical IR methods: Okapi BM25 and Latent Semantic Analysis (LSA). BM25 reduces the effect of very frequent words and handles document length better, while LSA helps match documents and queries with similar meaning but different words. Both methods improve the baseline in different ways.

We then combine BM25 and LSA into a hybrid model. Since each method fixes a different weakness of TF-IDF, the hybrid captures the strengths of both. Experiments show that the hybrid performs better than the baseline, BM25, and LSA on all evaluation metrics.

2 Problem Statement

The Cranfield search engine built in Parts 2–4 uses the normal Vector Space Model with TF-IDF weighting and cosine similarity. It is simple, transparent, and cheap to run, but it inherits two well-known weaknesses of the bag-of-words VSM that hold back its retrieval quality on realistic queries. The aim of Part 5 is to pin down these weaknesses, propose improvements based on established IR theory.

2.1 The Two Structural Assumptions of the Baseline

The TF-IDF Vector Space Model has two main weaknesses:

1. **Linear term frequency.** Repeating a word many times increases the score too much, even though extra repetitions may not add useful meaning. Longer documents are also unfairly favored [2].
2. **Independent terms.** The model treats every word separately, so it cannot understand synonyms or word meaning from context. Different words with similar meaning are missed, while the same word with different meanings can cause confusion [1].

2.2 Observed Failures on Cranfield

These limitations are not just theoretical. On Cranfield, the baseline reaches MAP@10 of 0.3122 and nDCG@10 of 0.4715 — respectable numbers that hide three distinct failure modes, which we see by inspecting individual queries (full case studies in Section 6):

- *Long, less relevant documents can rank too high.* For the query “wind-tunnel corrections for a two-dimensional aerofoil mounted off-centre in a tunnel,” the baseline retrieves only one of the three relevant documents in the top five. Instead, it ranks long papers higher simply because they repeatedly contain words like *aerofoil*, *tunnel*, and *corrections*.
- *Relevant documents using different words are often missed.* For the query “interaction between adjacent blade rows of a supersonic cascade,” the baseline misses documents describing the same idea with different wording and instead retrieves less relevant documents that only share similar terms.
- *Words with multiple meanings can confuse the model.* For the query “scale height in an atmosphere,” the baseline retrieves documents where *height* refers to something completely different, because the model cannot understand word meaning from context.

3 Background

This section reviews the three retrieval models that form the basis of our work: the Vector Space Model with TF-IDF weighting (the baseline from Parts 2–4),

Latent Semantic Analysis, and Okapi BM25. The goal is to make the theoretical motivation for each subsequent improvement explicit, so that the empirical comparisons in Section 5 can be interpreted in terms of which limitation of the baseline each method addresses.

3.1 Vector Space Model with TF-IDF

The Vector Space Model (VSM) represents documents and queries as vectors of terms. In TF-IDF, the weight of a term depends on how often it appears in a document and how rare it is across the collection:

$$\text{tfidf}(t, d) = \text{tf}(t, d) \cdot \log \frac{N}{\text{df}(t)}. \quad (1)$$

Documents are ranked using cosine similarity between query and document vectors. Although TF-IDF is simple and effective, it has two main weaknesses: repeated terms can influence the score too much, and words with similar meanings are treated as completely unrelated. These problems motivate the improved methods used later.

3.2 Latent Semantic Analysis

Latent Semantic Analysis (LSA) improves TF-IDF by matching documents based on meaning instead of exact words. It builds a TF-IDF term-document matrix and reduces it into a smaller latent space using truncated SVD:

$$M \approx U_K \Sigma_K V_K^\top, \quad (2)$$

where K is the number of latent dimensions.

In this space, words with similar meanings are grouped together, so queries and documents using different wording can still match. LSA also reduces noise by removing weaker patterns, though it can sometimes weaken exact keyword matches.

3.3 Okapi BM25

Okapi BM25 improves TF-IDF by reducing the effect of repeated terms and handling document length better [3, 2]. Instead of increasing scores linearly with term frequency, BM25 uses a saturation function:

$$\frac{\text{tf}(t, d) (k_1 + 1)}{\text{tf}(t, d) + k_1 \left(1 - b + b \cdot \frac{|d|}{\text{avgdl}}\right)}. \quad (3)$$

This means the first few occurrences of a term matter most, while repeated occurrences contribute less. BM25 also normalizes for document length, preventing long documents from getting unfairly high scores.

The full BM25 scoring formula is

$$\text{BM25}(d, q) = \sum_{t \in q} \log \frac{N - \text{df}(t) + 0.5}{\text{df}(t) + 0.5} \cdot \frac{\text{tf}(t, d) (k_1 + 1)}{\text{tf}(t, d) + k_1 \left(1 - b + b \cdot \frac{|d|}{\text{avgdl}}\right)}. \quad (4)$$

BM25 remains one of the strongest classical retrieval methods and is widely used in search engines such as Lucene and Elasticsearch.

3.4 Complementarity and Motivation for a Hybrid

LSA and BM25 improve different weaknesses of TF-IDF. LSA helps when queries and documents use different words with similar meaning, while BM25 improves ranking when repeated terms or long documents unfairly affect scores.

Because the two methods solve different problems, combining them can capture the strengths of both. We use a weighted hybrid score:

$$s_{\text{hybrid}}(d, q) = \alpha \cdot \tilde{s}_{\text{BM25}}(d, q) + (1 - \alpha) \cdot \tilde{s}_{\text{LSA}}(d, q), \quad (5)$$

where α controls the contribution of each method and $\tilde{s}$ represents normalized scores.

4 Approach

We propose three improvements over the baseline TF-IDF VSM from Parts 2–4: an LSA retriever, an Okapi BM25 retriever, and a hybrid that combines the two. All four systems — the baseline and the three new ones — share the same preprocessing pipeline and the same evaluation module from Parts 2–4, so any differences in retrieval quality come from the algorithms themselves, not from incidental differences in text processing or measurement.

4.1 Common Preprocessing Pipeline

Documents and queries are processed using the same preprocessing pipeline. Text is converted to lowercase, tokenized, filtered to keep only alphabetic words, stemmed using the Porter stemmer, and cleaned using the NLTK English stop-word list.

For TF-IDF and LSA, the processed tokens are joined into strings for `TfidfVectorizer`, while BM25 uses the token lists directly with `BM25Okapi`. Both representations contain the same information in different formats.

4.2 Baseline TF-IDF Retriever

The baseline replicates the system from Parts 2–4. The full term–document TF-IDF matrix is constructed according to Eq. 1, both document and query vectors are L2-normalized, and relevance is computed as cosine similarity. Documents are ranked in descending order of similarity for each query. This system serves as the reference point against which all proposed methods are statistically compared in Section 5.

4.3 LSA Retriever

The LSA retriever uses the TF-IDF matrix and applies truncated SVD (Eq. 2) to project documents and queries into a lower-dimensional latent space, where similarity is measured using cosine similarity.

The main parameter is the number of latent dimensions K . We tested $K \in \{50, 100, 150, 200, 300, 500\}$ and selected $K = 200$, which gave the best overall performance (Section 5.2).

We use `TruncatedSVD` from `scikit-learn` with `random_state=42` and normalize vectors before computing similarity.

4.4 BM25 Retriever

The BM25 retriever follows the standard Okapi form (Eq. 4) with $k_1 = 1.5$ and $b = 0.75$, the widely-used defaults from the BM25 literature [2] and the defaults in production systems like Lucene and Elasticsearch. We do not tune these on Cranfield, which avoids giving BM25 an unfair edge in the headline comparison.

We use the `rank_bm25` library. For each query, we score all 1400 documents and sort in descending order. Unlike TF-IDF and LSA, BM25 returns unbounded positive scores rather than cosine similarities in $[-1, 1]$ — a difference that matters for the hybrid system below.

4.5 Hybrid BM25–LSA Retriever

The hybrid retriever combines BM25 and LSA scores using Eq. 5. Since BM25 and LSA scores are on different scales, we first apply per-query min-max normalization:

$$\tilde{s}_q(d) = \frac{s_q(d) - \min_{d'} s_q(d')}{\max_{d'} s_q(d') - \min_{d'} s_q(d')}. \quad (6)$$

The mixing weight $\alpha \in [0, 1]$ controls the balance between BM25 and LSA. We tested values from 0.0 to 1.0 and selected $\alpha = 0.5$, which gave the best overall results.

5 Experimentation

the section is about the evaluation of 4 retrieval systems from Section 4. We first describe the dataset and evaluation setup (Section 5.1), then justify the two hyperparameter choices — K for LSA and α for the hybrid — via systematic sweeps (Sections 5.2, 5.3). We then report the headline four-way comparison across all six metrics (Section 5.4) and test each pairwise difference for using paired Wilcoxon signed-rank tests (Section 5.6).

5.1 Dataset and Evaluation Setup

Dataset. We evaluate on the Cranfield collection, which contains 1400 aerodynamics documents, 225 queries, and graded relevance judgments. For binary metrics, all relevance levels are treated as relevant; for nDCG, graded relevance scores are used.

Metrics. We evaluate all systems using Precision@ k , Recall@ k , F0.5@ k , MAP@ k , nDCG@ k , and MRR@ k for $k \in \{1, \dots, 10\}$. The same evaluation pipeline is used for all systems to ensure fair comparison.

Hyperparameters. All systems use the same preprocessing pipeline. LSA uses $K = 200$, BM25 uses $k_1 = 1.5$ and $b = 0.75$, and the hybrid uses $\alpha = 0.5$.

5.2 Hyperparameter Sweep: LSA Dimensionality K

To pick the latent dimensionality K , we evaluated LSA at $K \in \{50, 100, 150, 200, 300, 500\}$, scoring each setting on all six metrics at $k = 10$. Results are in Table 1 and Figure 1.

Table 1: LSA performance vs. latent dimensionality K , evaluated at $k = 10$ on Cranfield. Bold marks the best value per column.

K	P@10	R@10	F0.5@10	MAP	nDCG@10	MRR
50	0.2516	0.3628	0.2567	0.2443	0.3885	0.6232
100	0.2902	0.4172	0.2961	0.3061	0.4555	0.7050
150	0.2996	0.4296	0.3056	0.3259	0.4779	0.7478
200	0.3036	0.4351	0.3093	0.3280	0.4808	0.7432
300	0.2991	0.4302	0.3052	0.3226	0.4810	0.7551
500	0.2942	0.4241	0.3000	0.3180	0.4799	0.7685

The curve has the shape you would expect for LSA on a small corpus. At $K = 50$ the space is too compressed — MAP drops to 0.2443, even *below* the baseline of 0.3122 — while at $K \geq 300$ the extra low-singular-value components just dilute structure without helping. $K = 200$ gives the best P@10, R@10, F0.5@10, and MAP, and comes within 0.0003 of the best nDCG@10 and within 0.025 of the best MRR. We therefore use $K = 200$ for all later experiments.

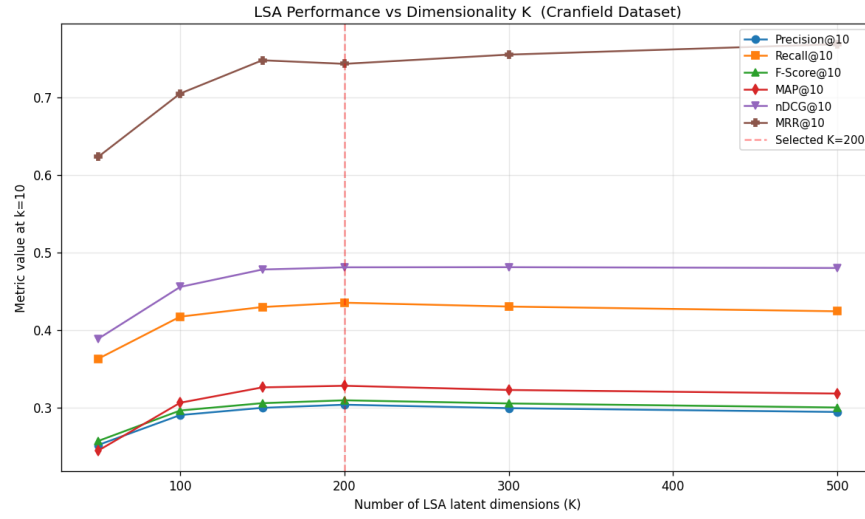


Fig. 1: LSA performance vs. latent dimensionality K on Cranfield. All six metrics rise sharply from $K = 50$, plateau in the $K = 150$ – 300 range, and either stay flat or drift slightly for $K \geq 300$. The selected $K = 200$ is marked by the dashed vertical line.

5.3 Hyperparameter Sweep: Hybrid Mixing Weight α

To select the mixing weight α for the hybrid, we evaluated 11 settings $\alpha \in \{0.0, 0.1, \dots, 1.0\}$, where $\alpha = 0.0$ corresponds to pure LSA and $\alpha = 1.0$ to pure BM25. Results are shown in Table 2 and Figure 2.

Table 2: Hybrid performance vs. mixing weight α , evaluated at $k = 10$ on Cranfield. $\alpha = 0.0$ is pure LSA, $\alpha = 1.0$ is pure BM25. Bold marks the best value per column.

α	P@10	R@10	F0.5@10	MAP	nDCG@10	MRR
0.0	0.3036	0.4351	0.3093	0.3280	0.4808	0.7432
0.1	0.3084	0.4428	0.3141	0.3380	0.4938	0.7662
0.2	0.3107	0.4474	0.3167	0.3446	0.5008	0.7771
0.3	0.3084	0.4453	0.3144	0.3487	0.5033	0.7870
0.4	0.3116	0.4513	0.3178	0.3543	0.5104	0.7950
0.5	0.3142	0.4544	0.3204	0.3578	0.5159	0.7987
0.6	0.3124	0.4505	0.3185	0.3583	0.5147	0.7948
0.7	0.3080	0.4456	0.3143	0.3529	0.5105	0.7934
0.8	0.3022	0.4393	0.3087	0.3467	0.5047	0.7862
0.9	0.2960	0.4308	0.3025	0.3420	0.4992	0.7781
1.0	0.2951	0.4333	0.3023	0.3390	0.4973	0.7776

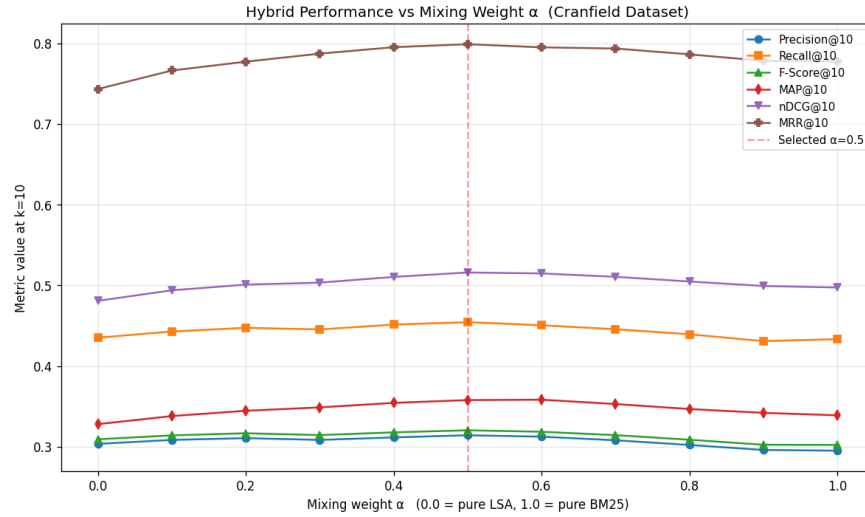


Fig. 2: Hybrid performance for different mixing weights α on Cranfield. Performance peaks near $\alpha = 0.5$, showing that combining BM25 and LSA works best. The endpoints $\alpha = 0.0$ and $\alpha = 1.0$ reproduce the standalone LSA and BM25 results, confirming the implementation is correct.

The results show three key points. First, $\alpha = 0.0$ reproduces pure LSA and $\alpha = 1.0$ reproduces pure BM25, confirming the implementation is correct. Second, performance peaks near $\alpha = 0.5$, supporting the idea that BM25 and LSA complement each other. Third, the hybrid at $\alpha = 0.5$ performs better than both LSA and BM25 on all metrics. Therefore, we choose $\alpha = 0.5$ for the final hybrid model.

5.4 Headline Four-Way Comparison

Table 3 reports each system’s mean score across all 225 Cranfield queries at the standard depth $k = 10$. All four share the preprocessing pipeline of Section 4.1 and are scored by the same **Evaluation** class.

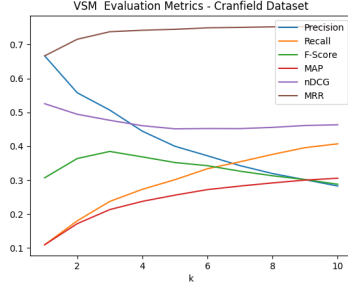
Table 3: Four-way comparison at $k = 10$ on Cranfield (225 queries). The Hybrid system wins on every metric. Best per column in bold.

System	P@10	R@10	F0.5@10	MAP	nDCG@10	MRR
Baseline TF-IDF	0.2853	0.4115	0.2909	0.3122	0.4715	0.7692
LSA ($K = 200$)	0.3036	0.4351	0.3093	0.3280	0.4808	0.7432
BM25 ($k_1=1.5, b=0.75$)	0.2951	0.4333	0.3023	0.3390	0.4973	0.7776
Hybrid ($\alpha = 0.5$)	0.3142	0.4544	0.3204	0.3578	0.5159	0.7987

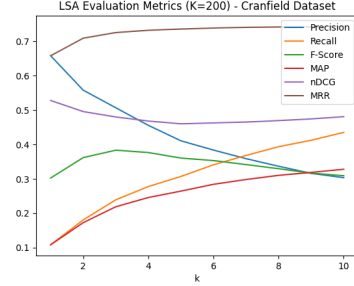
The hybrid model gives the best results on all six metrics, with the largest improvements on MAP and nDCG@10. LSA and BM25 both improve over the baseline, but in different ways: LSA performs better on precision-based metrics, while BM25 performs better on ranking metrics such as MAP, nDCG, and MRR. Although LSA lowers MRR compared to the baseline, the hybrid recovers this and achieves the highest MRR overall, combining the strengths of both methods.

5.5 Per-System Metric Curves

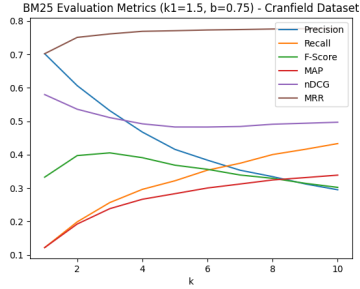
Figure 3 shows the six evaluation metrics as a function of $k \in \{1, \dots, 10\}$ separately for each of the four systems. The shape of each curve is informative beyond what the single $k = 10$ snapshot of Table 3 can convey.



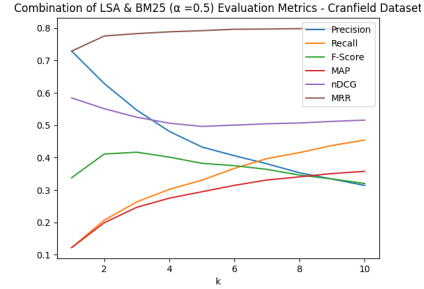
(a) Baseline TF-IDF.



(b) LSA ($K = 200$).



(c) BM25 ($k_1 = 1.5, b = 0.75$).



(d) Hybrid ($\alpha = 0.5$).

Fig. 3: All six evaluation metrics as a function of k for the four retrieval systems on Cranfield. The hybrid model achieves the best overall performance, especially on MAP and nDCG. Across all systems, precision decreases with k , recall increases with k , and F0.5-score peaks near $k = 3$.

The plots show three main patterns. First, all systems follow the same general behavior: precision decreases with k , recall increases, and F0.5-score peaks near

$k = 3$. Second, BM25 and the hybrid achieve better top-rank precision and nDCG@1 than the baseline. Third, the hybrid performs best overall, with almost all curves staying above the other systems, showing the benefit of combining BM25 and LSA.

5.6 Hypothesis Tests

Average metric values can hide how systems behave on individual queries, so we compare systems query by query using paired statistical tests.

Test setup. For each system pair and metric, we use per-query scores from the **Evaluation** module and apply paired Wilcoxon signed-rank tests, along with paired t -tests as a cross-check. Effect size is measured using Cohen’s d . Significance levels are marked as * for $p < 0.05$, ** for $p < 0.01$, and *** for $p < 0.001$.

Each method vs. baseline. Table 4 reports each of the three proposed methods against the baseline.

The hybrid model significantly improves all six metrics over the baseline, with five metrics at $p < 0.001$. LSA mainly improves precision-based metrics, while BM25 mainly improves ranking metrics such as AP and nDCG. Their different strengths support the idea that BM25 and LSA complement each other.

Hybrid vs. each constituent. Table 5 verifies that the hybrid is genuinely better than either method alone, not merely a successful score interpolation.

All twelve differences are statistically significant. Hybrid versus LSA shows particularly large effects on the metrics where LSA was weak (RR, nDCG, MAP), and Hybrid versus BM25 shows large effects on the metrics where BM25 was weak (P, R, F0.5). This is exactly the pattern expected if the hybrid is genuinely capturing both methods’ strengths simultaneously.

LSA vs. BM25. For completeness, Table 6 compares the two individual improved methods.

There is no statistically significant difference between LSA and BM25 overall. However, LSA performs slightly better on precision and recall metrics, while BM25 performs better on ranking metrics such as AP, nDCG, and RR. This shows that the two methods complement each other, which explains why the hybrid performs best overall.

5.7 Summary of Findings

The four-way comparison and hypothesis tests together establish three formal claims, each of the form prescribed by the assignment:

Table 4: Paired tests against baseline TF-IDF at $k = 10$. Δ is the mean difference (positive favors the proposed method). All p -values are paired Wilcoxon. Cohen’s d is computed on the paired differences.

Metric	LSA vs Baseline			BM25 vs Baseline			Hybrid vs Baseline		
	Δ	p	d	Δ	p	d	Δ	p	d
P	+0.018	0.003**	+0.20	+0.010	0.190	+0.10	+0.029	4.0×10^{-6} ***	+0.33
R	+0.024	0.010*	+0.18	+0.022	0.040*	+0.15	+0.043	6.7×10^{-6} ***	+0.32
F0.5	+0.018	0.007**	+0.20	+0.011	0.171	+0.12	+0.029	2.4×10^{-6} ***	+0.33
AP	+0.016	0.125	+0.15	+0.027	0.002**	+0.21	+0.046	5.7×10^{-9} ***	+0.39
nDCG	+0.009	0.317	+0.09	+0.026	0.001**	+0.18	+0.044	4.9×10^{-8} ***	+0.37
RR	-0.026	0.050	-0.12	+0.008	0.606	+0.03	+0.030	0.046*	+0.13

Table 5: Paired tests of Hybrid against each constituent method at $k = 10$. The hybrid significantly improves over both LSA and BM25 on every metric.

Metric	Hybrid vs LSA			Hybrid vs BM25		
	Δ	p	d	Δ	p	d
P	+0.011	0.035*	+0.16	+0.019	0.001***	+0.23
R	+0.019	0.043*	+0.15	+0.021	0.008**	+0.19
F0.5	+0.011	0.022*	+0.16	+0.018	0.001***	+0.22
AP	+0.030	1.0×10^{-4} ***	+0.30	+0.019	0.002**	+0.21
nDCG	+0.035	2.3×10^{-5} ***	+0.31	+0.019	0.005**	+0.21
RR	+0.056	2.1×10^{-4} ***	+0.24	+0.021	0.043*	+0.11

1. The hybrid system ($\alpha \cdot \text{BM25} + (1 - \alpha) \cdot \text{LSA}$ at $\alpha = 0.5$, with LSA $K = 200$ and BM25 $k_1 = 1.5, b = 0.75$) is better than the baseline TF-IDF VSM with respect to every one of the six evaluated metrics on Cranfield at $k = 10$, with paired Wilcoxon $p < 0.05$ minimum and $p < 0.001$ on five of the six metrics (P, R, F0.5, AP, nDCG).
2. The hybrid system is better than each of its constituent methods (LSA alone and BM25 alone) with respect to every metric, with paired Wilcoxon $p < 0.05$ minimum, confirming that the combination is a genuine improvement rather than an interpolation.
3. LSA alone and BM25 alone improve over the baseline through statistically distinguishable channels: LSA wins significantly on precision-oriented metrics (P, R, F0.5), BM25 wins significantly on aggregate ranking metrics (AP, nDCG), and the two are statistically indistinguishable in head-to-head comparison ($p > 0.085$ throughout). This complementarity is the empirical basis for the hybrid’s success.

The interpretation of these claims at the level of individual query failures — which the aggregate metrics smooth over — is taken up in the next section.

Table 6: Paired tests of BM25 against LSA at $k = 10$. No metric shows a statistically significant difference between the two methods.

Metric	Δ	Wilcoxon p	Cohen's d
P	-0.008	0.194	-0.08
R	-0.002	0.583	-0.01
F0.5	-0.007	0.245	-0.06
AP	+0.011	0.527	+0.07
nDCG	+0.017	0.168	+0.10
RR	+0.034	0.086	+0.11

6 Discussion

The results in Section 5 show that the hybrid system significantly improves over the baseline and that LSA and BM25 help in different ways. In this section, we examine specific Cranfield queries where the baseline performs poorly and the hybrid performs better, helping us understand the main weaknesses of the Vector Space Model.

6.1 Methodology for Failure Analysis

For all 225 Cranfield queries, we compared the AP@10 scores of the hybrid and baseline systems and selected the queries with the largest improvements. For each query, we examined the query text, AP@10 scores, relevant documents, and top retrieved documents.

The analysis reveals three main failure patterns in the baseline system. Some are mainly fixed by BM25 through better term weighting and length normalization, while others are mainly fixed by LSA through semantic matching and handling word ambiguity.

The first failure mode happens when long documents repeat query terms many times even though they are not truly relevant. The baseline gives such documents high scores because it treats term frequency linearly.

Case 1 (Query 81) asks about “*wind-tunnel corrections for a two-dimensional aerofoil mounted off-centre in a tunnel.*” The relevant documents are 631, 672, and 799. The baseline retrieves only one relevant document in the top five and instead ranks unrelated long documents such as 249 and 714, which repeatedly contain words like *aerofoil*, *tunnel*, and *corrections*. BM25 fixes this problem and achieves $AP@10 = 1.000$, compared to the baseline’s 0.331, by correctly ranking all three relevant documents in the top three positions.

Case 2 (Query 119) asks about “*axisymmetric deviations from circularity*” in “*cylinders under hydrostatic pressure.*” The relevant documents are 897 and 926. The baseline misses document 897 and instead retrieves unrelated cylinder documents such as 789 and 1133. BM25 ranks document 897 first and achieves $AP@10 = 0.833$, while LSA fails completely on this query with $AP@10 = 0.000$. This shows that BM25 helps where LSA’s semantic smoothing can hurt.

6.2 Failure Mode 2: Synonymy and Topical Paraphrase

The second failure mode occurs when relevant documents discuss the same topic as the query but use different words. The baseline depends on exact word matching, while LSA can match related meanings.

Case 4 (Query 25) asks about “*the interaction between adjacent blade rows of a supersonic cascade.*” There are 10 relevant documents. The baseline retrieves only the most directly matching ones (IDs 277, 214) and then returns less relevant documents such as 772 and 990 because they share words like *blade* and *flow*. LSA improves AP@10 from 0.300 to 0.716 by retrieving document 511, which discusses the same idea using different wording (*double cascade rotor-nozzle interaction*).

Case 3 (Query 208) asks about “*the shape of the drag polar of a lifting spacecraft*” and reducing deceleration during re-entry. Out of 7 relevant documents, the baseline retrieves only two and then returns unrelated documents such as 566 and 1124 because they share words like *drag*. LSA achieves AP@10 = 0.800 by retrieving documents from the correct re-entry topic cluster, including document 163, whose title shares almost no query words.

6.3 Failure Mode 3: Polysemy and Term Overloading

The third failure mode occurs when a query word has multiple meanings. The baseline cannot distinguish the correct meaning and retrieves documents from the wrong context.

Case 5 (Query 180) asks: “*how does scale height vary with altitude in an atmosphere?*” There are 8 relevant documents. The baseline retrieves some correct documents but also returns document 314, where *height* refers to the height of roughness particles in boundary-layer theory, not atmospheric physics. LSA achieves AP@10 = 0.947 by ranking all 8 relevant documents highly, including documents discussing related ideas such as *air density variations* and *atmospheric drag*. This shows that LSA can better capture the correct meaning of words from context.

6.4 Why the Hybrid Wins on All Three Modes

The five cases show that BM25 and LSA improve different weaknesses of the baseline. BM25 fixes problems caused by long, term-heavy documents (Cases 1 and 2), while LSA fixes vocabulary mismatch and word ambiguity (Cases 3, 4, and 5). In some cases, such as Query 119, LSA even hurts performance while BM25 helps.

The hybrid succeeds because it combines the strengths of both methods. A document can rank higher if either BM25 or LSA scores it well. This also explains why the hybrid achieves the best MRR (0.7987), outperforming even BM25 alone (0.7776).

6.5 What This Tells Us About the VSM's Limitations

The failure analysis shows that the baseline Vector Space Model has three different weaknesses. Although the overall MAP improvement is only 0.046, it comes from fixing different types of retrieval problems, such as poor term weighting and vocabulary mismatch. BM25 and LSA address these weaknesses in different ways, and combining them with a simple hybrid score gives better performance than either method alone. This shows that the baseline problems are structural and cannot be solved by simple parameter tuning.

7 Conclusion and Future Work

This report improved the baseline TF-IDF Vector Space Model from Parts 2–4 of the Cranfield search-engine project. We evaluated three methods: LSA with $K = 200$, BM25 with $k_1 = 1.5$ and $b = 0.75$, and a hybrid model that combines BM25 and LSA using per-query normalization and $\alpha = 0.5$.

7.1 Summary of Findings

Three formal claims emerge from the experimental evidence, all in the form prescribed by the project:

- The hybrid system is statistically better than the baseline TF-IDF VSM on Cranfield at $k = 10$ with respect to every one of the six evaluated metrics, with paired Wilcoxon $p < 0.05$ on Mean Reciprocal Rank and $p < 0.001$ on Precision, Recall, F-score, Mean Average Precision, and normalized Discounted Cumulative Gain, under the assumptions of $K = 200$, $k_1 = 1.5$, $b = 0.75$, and $\alpha = 0.5$.
- The hybrid system is statistically better than the baseline TF-IDF VSM on Cranfield at $k = 10$ with respect to every one of the six evaluated metrics, with paired Wilcoxon $p < 0.05$ on Mean Reciprocal Rank and $p < 0.001$ on Precision, Recall, F0.5-score, Mean Average Precision, and normalized Discounted Cumulative Gain, under the assumptions of $K = 200$, $k_1 = 1.5$, $b = 0.75$, and $\alpha = 0.5$.
- LSA and BM25 alone are statistically indistinguishable in head-to-head comparison ($p > 0.085$ throughout), but the metrics on which each shows the strongest gains against the baseline are different: LSA significantly improves the precision-oriented family (P, R, F) while BM25 significantly improves the ranking-quality family (AP, nDCG). This complementarity is the empirical mechanism by which the hybrid succeeds where neither method alone would.

The hybrid model shows clear improvement over the baseline. It achieves $\text{MAP} = 0.3578$ compared to the baseline's 0.3122 (a 14.6% improvement) and $\text{nDCG}@10 = 0.5159$ compared to 0.4715 (+9.4%).

Query analysis revealed three main problems in the baseline system: repeated terms in long irrelevant documents, different words used for the same meaning, and words with multiple meanings. BM25 and LSA fix these problems in different ways, which explains the strong performance of the hybrid model.

7.2 Future Work

Broader retrieval methods. The hybrid framework can be extended by adding more retrieval methods, such as Explicit Semantic Analysis (ESA) or modern neural retrievers. The same score-normalization approach can combine multiple systems together.

Query-side improvements. Future work can improve query handling through spelling correction, query auto-completion, and query expansion methods such as pseudo-relevance feedback or word embeddings.

Larger benchmarks. Cranfield is small and focused only on aerodynamics. Testing the hybrid on larger datasets such as TREC or MS MARCO would show whether the same parameter choices and improvements generalize to other domains.

Efficiency considerations. The current implementation focuses on correctness rather than speed. Faster indexing, approximate nearest-neighbor search, and optimized BM25 implementations could improve efficiency without changing retrieval quality.

Final remarks. The main conclusion is that the weaknesses of TF-IDF VSM are structural, not just parameter-related. BM25 and LSA improve retrieval because they add information that simple term-overlap scoring cannot capture.

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